Vaibhav Kumar

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.00
Vardhman Senior Secondary School Suket	Class XII (RBSE)	2022	96.60%
Govt Senior Secondary School Suket	Class X (RBSE)	2020	97.33%

EXPERIENCE

Expedia Group India

Software Development Engineering Intern

- Developed **8 automated regression specs** for critical Business Traveler (HBT) journeys using **WebdriverIO, Cucumber & Page Object Model**.
- Automated **Intent Capture and Business Highlights** across search results and property pages.
- Covered **6 user interaction paths** including popup/banner Yes, No, close and tap-outside flows.
- Implemented **TNL force-bucketing** for deterministic experiment-based testing.
- Built a **2,400-combination input rotation utility** with file locking and 7-day no-repeat logic to reduce test flakiness.
- Integrated HBT tests with **GitHub Actions**, Allure reporting and **Slack notifications** for CI validation.
- Debugged CI/pipeline issues and used **AI-assisted development** for scaffolding, debugging and test-quality improvements.

SKILLS

Programming Languages - C++, C, Python, SQL, C++(OOPs proficient), Kotlin.

Libraries/Frameworks - Jetpack Compose, NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, StatsModels.

Tools - Git, GitHub, Jupyter Notebook, Google Colab, VS Code, Anaconda, Android Studio

Concepts - Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), Operating System, DBMS

PROJECTS

Quantum Annealing | Python, NumPy, SciPy, Matplotlib

- Built a numerical **quantum annealing simulator** for a 5-spin Ising model using matrix-based computational methods.
- Implemented **Hamiltonian construction, exact diagonalization, RK4 integration, and matrix-exponential time evolution**.
- Performed **convergence and numerical stability analysis** across varying time steps and annealing durations.
- Evaluated performance using **ground-state overlap and residual energy**, with visualizations of simulation results.

Interactive Android Math Game - *Jetpack Compose*

- Built an interactive Android Math Game using **Jetpack Compose modern UI using Material Design 3 component.**
- Implemented real-time game mechanics with 30-second timer, scoring system (10 points per correct answer), and dynamic state management.
- Utilized **Jetpack Navigation** for seamless screen transitions and argument passing between **composables.**
- Developed in **Kotlin** leveraging modern Android development best practices and declarative UI programming paradigms.

Home Devices Management System - C++

- Developed a Home Device Management System to address **daily smart home automation needs**, enabling secure device control for **multiple user roles.**
- Implemented robust backend logic in **C++ utilizing Object-Oriented Programming** principles such as **abstraction**, **inheritance**, and **polymorphism**.

RELEVANT COURSES

- **Computer / AI & ML** - Data Structure and Application, Computational Physics, Introduction to Artificial Intelligence, Introduction to Programming
- **Mathematics** - Advanced Mathematical Physics, Introduction to Probability, Introduction to Statistics, Differential Equation, Calculus - I, Calculus - II, Linear Algebra
- **Physics** - Optics and Photonics, Special Relativity, Electronic Device Physics, Classical Mechanics
- **Other** - Analog Electronics, Digital Electronics, Introduction to Entrepreneurship, Basic Electrical Engineering

EXTRACURRICULAR

1. Participated in Inter-IIT Cultural Meet 7.0 held at IIT Patna; represented IIT Hyderabad in Street Play and Hindi Creative Writing events.
2. Actively involved in dramatics and literary activities on campus, contributing to cultural engagement and creative expression.
3. Member of the National Service Scheme (NSS), IIT Hyderabad - Completed 120+ hours

POSITIONS OF RESPONSIBILITY

Placement Coordinator

- Helped students with placement-related queries and coordinated with companies for placement drives

Event Coordinator

- Organized events, managed participants, and promoted club activities.

Internship Coordinator

- Helped students with internship-related queries and coordinated with companies for internship drives.